Identified Risk

Environment

- Server: Task execution failure (E01).
- Connection: Connection loss (E02).
- Web:
  - Connection loss in HTTP requests (E03).
  - Connection loss in HTTPS requests (E04).
- FTP: Task execution failure in transaction management (E05).
- SQL:
  - Failure in content access (E15).
  - Failure in permission granting (E06).
- Task Execution:
  - Failure in asynchronous task execution (E07).
  - Failure in asynchronous task execution (E08).
- Data: Data loss and failure (E09).
- Integrity:
  - Loss of data stored on servers (E10).
  - Failure in access control rules (E11).
- Durability: Failure in data durability state (E12).
- Availability:
  - Failure in data exposure (E13).
  - Failure in data access (E14).

Architecture

- Security:
  - Session management failure (A01).
  - Input validation failure (A02).
- Interface:
  - Authentication failure (A03).
  - Authorization failure (A04).
- User Authentication: Loss of authentication in restricted areas (A05).
- Encryption: Cryptographic mechanism error (A06).
- Digital Certificates: Digital certificate mechanism failure (A07).
- Tokens: Security failure in token decoding (A08).

- Data Availability:
  - Persistence: Security failure in offline persistence (A09).
  - Data Integrity: Data security failure (A10).
- Efficiency:
  - Performance mechanism failure (A11).
  - Access mechanism failure (A12).
- Performance:
  - Resource contention failure (A13).
  - Cache usage failures (A14).
- Concurrent Access:
  - Data scalability failure (A5).
  - Deadlock prevention failure (A15).
  - Low performance in socket usage (A16).
- Operational Performance:
  - Connection failure when viewing media (A19).
  - Size: Data handling failure (A20).
  - Table Size: Failure in materialized view mechanism (A21).
  - Restricted Memory Usage: Slow operations with aligned vectors (A22).
  - Cache Necessity: Cache mechanism failure (A23).

Integration

- Legacy:
  - Documentation unavailability (I01)
  - Technological obsolescence (I02)
  - Complex dependencies (I03)
- Functionality Access: API compatibility failure (I04)
- Communication Methods: Low performance in microservices communication (I05)
- Data Exchange:
  - Data compliance failure (I06)
  - Real-time data exchange failure (I07)
- Third-Party:
  - Plugin integration failure (I08)
  - Notification cycle failures (I09)
  - Services: Failure with hotspots via third-party applications (I10)
  - Performance degradation (I11)
  - APIs: Loss of versioning configuration (I12)
  - Frameworks:

    - Framework compatibility failure (I13)
    - Framework versioning failures (I14)
  - Hybrid Development:
    - Compatibility failure (I15)
    - Versioning failure (I16)
    - Performance failure when using the framework (I17)
  - Ionic:
    - Compatibility failure with Ionic (I18)
    - Versioning failure in Ionic (I19)
  - Electron:
    - Compatibility failure with Electron (I20)
    - Versioning failure in Electron (I21)
  - Security:
    - Session authentication failure (I22)
    - Session management failure (I23)
  - JWT:
    - JWT validation failure (I24)
    - JWT signature failure (I25)
    - JWT status verification mechanism failure (I26)
  - Authentication:
    - Authentication mechanism failure (I27)
    - Failure in storing authentication credentials (I28)
  - Configurations:
    - Dependency failure (I29)
    - Versioning configuration failure (I30)
  - Libraries:
    - Library configuration error (I31)
    - Library dependency failure (I32)
  - Requests:
    - SOAP configuration loss (I33)
    - Response validation failure (I34)
  - Persistence:
    - Error handling failure (I35)
    - Security patch versioning failure (I36)
  - Communication:
    - UDP library configuration failure (I37)
    - TCP library configuration failure (I38)
    - Two-Phase Commit (TOC) error (I39)

Method

- Technique: Data handling failure (M01)
- Training Predictive Models:

  - Overfitting failure (M02)
  - Underfitting failure (M03)
  - Incorrect parameter selection (M04)
- Coding a Blockchain Network:
  - Security mechanism failure (M05)
  - Scalability handling failure (M06)
- Public: Consensus mechanism failure (M07)
- Private: Permission mechanism failure (M08)
- Consortium: Interoperability mechanism failure (M09)
- Implementing an AI Algorithm:
  - Hyperparameter tuning failure (M10)
  - Low algorithm performance (M11)
- Validation:
  - Incorrect understanding of requirements (M12)
  - Lack of communication and collaboration between the team and the client (M13)
- Client Data Requirement:
  - Data unavailability due to privacy concerns (M14)
  - Inaccurate client data (M15)
- System Deployment:
  - Low build performance (M16)
  - Driver update failure (M17)